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Answer

Of course! Let's break down this problem. You've got the right answer, which is fantastic! Now, let's understand the 'why' behind it, as this is a very important concept for JEE. ### **Analysis of the Question** * **Topic:** Relations & Functions (specifically, Equivalence Relations). * **Type:** **Conceptual / Counting Problem**. It tests your fundamental understanding of equivalence relations and a clever way to count them. --- ### **Method 1: The Partition Method (Highly Recommended)** This is the most elegant and efficient way to solve this problem. **The Golden Rule:** The number of equivalence relations on a set is equal to the number of ways you can **partition** that set. **What is a partition?** A partition of a set is just a way of splitting it into non-empty, non-overlapping subsets (think of them as "boxes") that together make up the whole set. Let our set be `A = {1, 2, 3}`. Let's find all the ways we can partition these three elements. **Case 1: All elements in one box.** * **Visual:** `[Box 1: {1, 2, 3}]` * **Partition:** `{{1, 2, 3}}` * **Number of ways:** 1 * *This corresponds to the largest equivalence relation (the Universal Relation) where every element is related to every other element.* **Case 2: One element in one box, and the other two in a second box.** * We need to choose which element gets to be alone. * Let '1' be alone: `{{1}, {2, 3}}` * Let '2' be alone: `{{2}, {1, 3}}` * Let '3' be alone: `{{3}, {1, 2}}` * **Visuals:** * `[Box 1: {1}] [Box 2: {2, 3}]` * `[Box 1: {2}] [Box 2: {1, 3}]` * `[Box 1: {3}] [Box 2: {1, 2}]` * **Number of ways:** 3 * *For example, the partition {{1}, {2, 3}} corresponds to the relation R = {(1,1), (2,2), (3,3), (2,3), (3,2)}.* **Case 3: Each element in its own separate box.** * **Visual:** `[Box 1: {1}] [Box 2: {2}] [Box 3: {3}]` * **Partition:** `{{1}, {2}, {3}}` * **Number of ways:** 1 * *This corresponds to the smallest equivalence relation (the Identity Relation) where each element is only related to itself: R = {(1,1), (2,2), (3,3)}.* **Summing them up:** Total number of partitions = (Ways from Case 1) + (Ways from Case 2) + (Ways from Case 3) Total = 1 + 3 + 1 = **5** So, there are **5** possible partitions, which means there are **5** equivalence relations on the set {1, 2, 3}. **Awesome job on getting this right! It shows you have good intuition.** --- ### **Method 2: Building the Relations Manually** This method is slower but helps reinforce the definitions of Reflexive, Symmetric, and Transitive (RST). An equivalence relation must be **Reflexive**. This is our non-negotiable starting point. So, for the set `{1, 2, 3}`, any equivalence relation *must* contain the pairs `{(1,1), (2,2), (3,3)}`. 1. **The Smallest Relation (R1):** `R1 = {(1,1), (2,2), (3,3)}` This is reflexive, symmetric, and transitive. (This is our Case 3 from above). 2. **Adding one symmetric pair:** Let's add `(1,2)`. For symmetry, we must also add `(2,1)`. `R2 = {(1,1), (2,2), (3,3), (1,2), (2,1)}` This is RST. (Corresponds to partition `{{1, 2}, {3}}`) Similarly, we can form: `R3 = {(1,1), (2,2), (3,3), (1,3), (3,1)}` (Corresponds to `{{1, 3}, {2}}`) `R4 = {(1,1), (2,2), (3,3), (2,3), (3,2)}` (Corresponds to `{{2, 3}, {1}}`) This gives us 3 more relations. 3. **Adding more pairs:** Let's take R2 and add another pair, say `(2,3)`. We start with `{(1,1), (2,2), (3,3), (1,2), (2,1)}`. * Add `(2,3)`. For **symmetry**, we must add `(3,2)`. * Now we have `{(1,1), ..., (1,2), (2,1), (2,3), (3,2)}`. * For **transitivity**, we have `(1,2)` and `(2,3)`, so we *must* have `(1,3)`. * For **symmetry** again, if we add `(1,3)`, we *must* add `(3,1)`. * This forces us to include all possible pairs. `R5 = {(1,1), (2,2), (3,3), (1,2), (2,1), (1,3), (3,1), (2,3), (3,2)}` This is the Universal Relation. (This is our Case 1 from above). **Counting them:** We have R1, R2, R3, R4, and R5. That's a total of **5** relations. --- ### **Mentor's Advice & Key Takeaways** * **Bell Numbers:** The number of ways to partition a set with `n` elements is called the n-th **Bell Number**, denoted by B(n). What you just calculated is B(3) = 5. * B(1) = 1 * B(2) = 2 * B(3) = 5 * B(4) = 15 Remembering these first few Bell numbers can help you solve MCQs in seconds! * **Focus on the Partition Method:** For counting equivalence relations, always think in terms of partitions. It's faster and less prone to errors. * **Non-empty condition:** The question mentioned "non-empty" equivalence relations. On a non-empty set like {1,2,3}, an equivalence relation must be reflexive, so it must contain {(1,1), (2,2), (3,3)}. Therefore, it can never be empty. This part of the question was just to ensure you're paying attention. ### **Practice Problems** 1. **Try this yourself:** Find the number of equivalence relations on the set {1, 2, 3, 4}. (Hint: You are looking for B(4). Try to list all 15 partitions!) 2. What is the smallest and largest equivalence relation on the set {a, b, c}? 3. **PYQ Reference:** Look up JEE Main questions from previous years on "Relations and Functions". You'll often find questions about identifying types of relations (RST) or counting them. This is a recurring theme. 4. **NCERT:** Go through Chapter 1, "Relations and Functions," in your Class 12th textbook to solidify the definitions. You're on the right track. Keep breaking down problems to their core concepts like this, and you'll do great. Keep practicing

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Analysis of the Question

- **Topic:** Relations & Functions (specifically, Equivalence Relations).
- **Type:** **Conceptual / Counting Problem**. It tests your fundamental understanding of equivalence relations and a clever way to count them.
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Method 1: The Partition Method (Highly Recommended)

This is the most elegant and efficient way to solve this problem.

- *The Golden Rule:** The number of equivalence relations on a set is equal to the number of ways you can **partition** that set.

What is a partition?

A partition of a set is just a way of splitting it into non-empty, non-overlapping subsets (think of them as "boxes") that together make up the whole set.

Let our set be `A = {1, 2, 3}`. Let's find all the ways we can partition these three elements.

Case 1: All elements in one box.

- **Visual:** `[Box 1: {1, 2, 3}]`

• **Partition:** `{{1, 2, 3}}`

• **Number of ways:** 1

• *This corresponds to the largest equivalence relation (the Universal Relation) where every element is related to every other element.*

Case 2: One element in one box, and the other two in a second box.

• We need to choose which element gets to be alone.

• Let '1' be alone: `{{1}, {2, 3}}`

• Let '2' be alone: `{{2}, {1, 3}}`

• Let '3' be alone: `{{3}, {1, 2}}`

• **Visuals:**

• `[Box 1: {1}] [Box 2: {2, 3}]`

• `[Box 1: {2}] [Box 2: {1, 3}]`

• `[Box 1: {3}] [Box 2: {1, 2}]`

• **Number of ways:** 3

• *For example, the partition {{1}, {2, 3}} corresponds to the relation R = {(1,1), (2,2), (3,3), (2,3), (3,2)}.*

Case 3: Each element in its own separate box.

• **Visual:** `[Box 1: {1}] [Box 2: {2}] [Box 3: {3}]`

• **Partition:** `{{1}, {2}, {3}}`

• **Number of ways:** 1

• *This corresponds to the smallest equivalence relation (the Identity Relation) where each element is only related to itself: R = {(1,1), (2,2), (3,3)}.*

Summing them up:

Total number of partitions = (Ways from Case 1) + (Ways from Case 2) + (Ways from Case 3)

Total = 1 + 3 + 1 = **5**

So, there are **5** possible partitions, which means there are **5** equivalence relations on the set {1, 2, 3}.

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Method 2: Building the Relations Manually

This method is slower but helps reinforce the definitions of Reflexive, Symmetric, and Transitive (RST).

An equivalence relation must be **Reflexive**. This is our non-negotiable starting point. So, for the set `{1, 2, 3}`, any equivalence relation *must* contain the pairs `{(1,1), (2,2), (3,3)}`.

1. **The Smallest Relation (R1):**
`R1 = {(1,1), (2,2), (3,3)}`

This is reflexive, symmetric, and transitive. (This is our Case 3 from above).

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Let's add `(1,2)`. For symmetry, we must also add `(2,1)`.

`R2 = {(1,1), (2,2), (3,3), (1,2), (2,1)}`

This is RST. (Corresponds to partition `{{1, 2}, {3}}`)

Similarly, we can form:

`R3 = {(1,1), (2,2), (3,3), (1,3), (3,1)}` (Corresponds to `{{1, 3}, {2}}`)

`R4 = {(1,1), (2,2), (3,3), (2,3), (3,2)}` (Corresponds to `{{2, 3}, {1}}`)

This gives us 3 more relations.

3. **Adding more pairs:**
Let's take R2 and add another pair, say `(2,3)`.

We start with `{(1,1), (2,2), (3,3), (1,2), (2,1)}`.

• Add `(2,3)`. For **symmetry**, we must add `(3,2)`.
• Now we have `{(1,1), ..., (1,2), (2,1), (2,3), (3,2)}`.
• For **transitivity**, we have `(1,2)` and `(2,3)`, so we *must* have `(1,3)`.
• For **symmetry** again, if we add `(1,3)`, we *must* add `(3,1)`.
• This forces us to include all possible pairs.

`R5 = {(1,1), (2,2), (3,3), (1,2), (2,1), (1,3), (3,1), (2,3), (3,2)}`

This is the Universal Relation. (This is our Case 1 from above).

• *Counting them:** We have R1, R2, R3, R4, and R5. That's a total of **5** relations.

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Mentor's Advice & Key Takeaways

• **Bell Numbers:** The number of ways to partition a set with `n` elements is called the n-th **Bell Number**, denoted by B(n). What you just calculated is B(3) = 5.

• B(1) = 1

• B(2) = 2

• B(3) = 5

• B(4) = 15

Remembering these first few Bell numbers can help you solve MCQs in seconds!

• **Focus on the Partition Method:** For counting equivalence relations, always think in terms of partitions. It's faster and less prone to errors.

• **Non-empty condition:** The question mentioned "non-empty" equivalence relations. On a non-empty set like {1,2,3}, an equivalence relation must be reflexive, so it must contain {(1,1), (2,2), (3,3)}. Therefore, it can never be empty. This part of the question was just to ensure you're paying attention.

Practice Problems

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Ask your queries

Type your question here...

Ask

Analyze

Follow-ups:

- I didnt understand step 3
- Elaborate partition

Topic Status: Weak

The student is lost on both a specific step and a core vocabulary term ('partition'), indicating a fragile, foundational understanding of the material.

In this legal document, the term "partition" is not used in the traditional sense of dividing property. Instead, it's used in a mathematical context related to sets. Think of it like this: Imagine you have a group of things, in this case, the numbers {1, 2, 3}. A **partition** of this group is like dividing it into smaller, distinct boxes. Here's what makes it a partition: * **Every item must be in a box:** All the original numbers (1, 2, and 3) must end up in one of the boxes. * **No item can be in more than one box:** Each number can only belong to a single box. * **The boxes can't be empty:** Each box you create must have at least one item in it. The question is asking how many different ways you can divide the numbers {1, 2, 3} into these distinct, non-empty boxes. The "equivalence relations" mentioned are a mathematical way of describing these groupings. So, in this context, **partition means dividing a set into non-overlapping, non-empty subsets that, when combined, contain all the original elements.**